NutriPATH
INTEGRATIVE PATHOLOGY SERVICES
P: 1300 688 522
E: info@nutripath.com.au

BENJAMIN TRENGROVE
03-Oct-1989 Male

1 INTERNATIONAL PATIENT
MELBOURNE VIC 3000

-.I SCREEN
INTELLIGENT SCREENING
PO BOX 8441, ANGELO STREET
SOUTH PERTH WA 6151

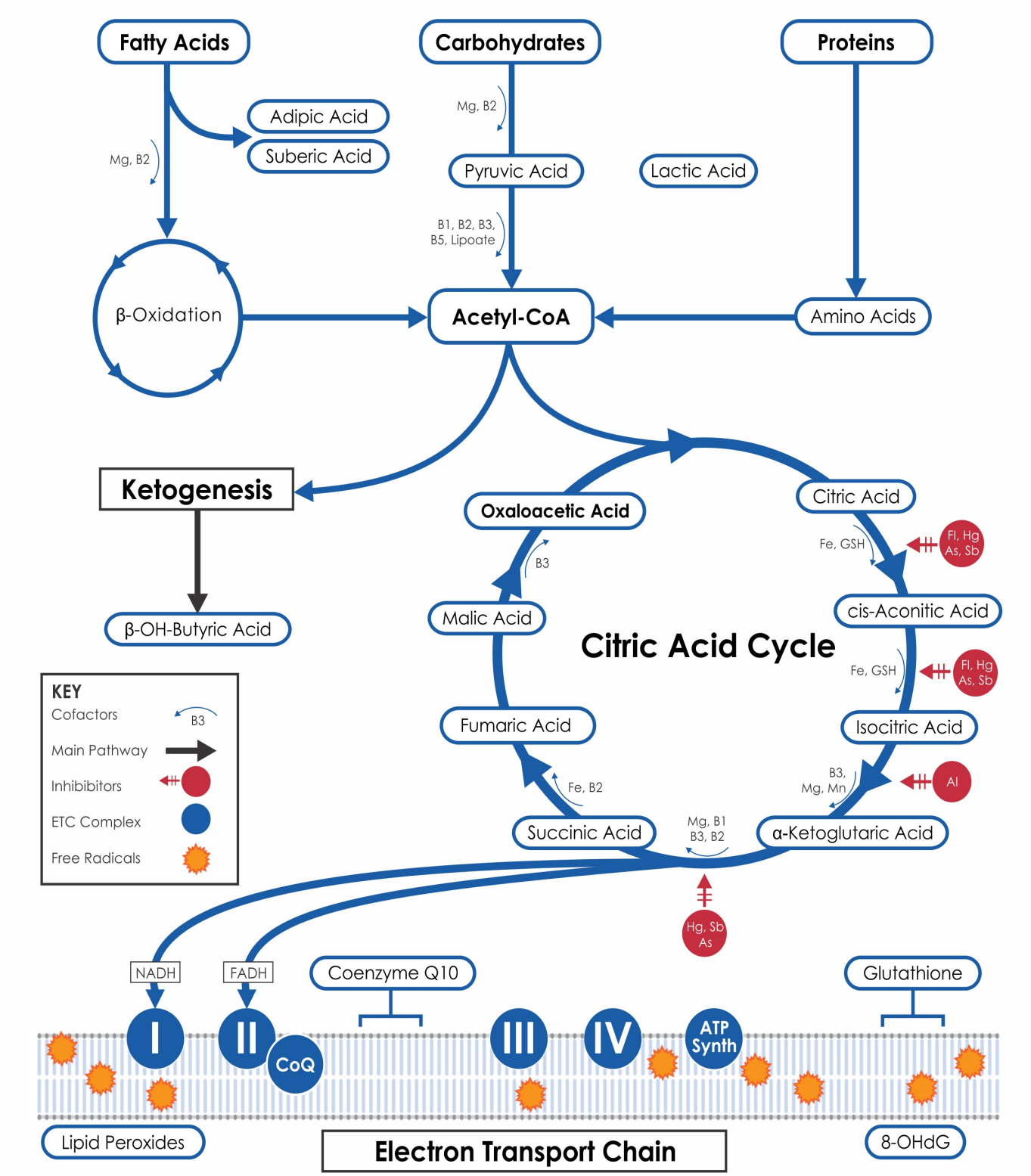
LAB ID : 3887019
UR NO. : 6190621
Collection Date : 23-Apr-2023
Received Date: 01-May-2023

3887019

ORGANIC ACIDS METABOLOMIC MAPPING

Method: LCMS/MS/MS

Organic Acids Pathways



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Nutrient Markers

URINE, SPOT

KETONE/FATTY ACID Metabolites
(Carnitine & B2)

KETONE/FATTY ACID Metabolites
(Carnitine & B2)

1.	Adipic Acid.	1.75	0.00 - 11.10	ug/mgCR
1.	Adipic Acid.	1.75	0.00 - 11.10	ug/mgCR
2.	Suberic Acid.	3.81	0.00 - 4.60	ug/mgCR
2.	Suberic Acid.	3.81	0.00 - 4.60	ug/mgCR
3.	Ethylmalonic Acid	1.34	0.00 - 6.30	ug/mgCR
3.	Ethylmalonic Acid	1.34	0.00 - 6.30	ug/mgCR
4.	Pimelic Acid	6.7	5.9 - 31.8	ug/mgCR
4.	Pimelic Acid	6.7	5.9 - 31.8	ug/mgCR
5.	Methyl-Succinic Acid	7.28	3.20 - 21.10	ug/mgCR
5.	Methyl-Succinic Acid	7.28	3.20 - 21.10	ug/mgCR

CARBOHYDRATE Metabolism/Glycolysis
(B1, B3, Cr, Lipoic Acid, CoQ10)

CARBOHYDRATE Metabolism/Glycolysis
(B1, B3, Cr, Lipoic Acid, CoQ10)

6.	Pyruvic Acid.	2.54	0.00 - 6.40	ug/mgCR
6.	Pyruvic Acid.	2.54	0.00 - 6.40	ug/mgCR
7.	Lactic Acid.	10.40	0.00 - 16.40	ug/mgCR
7.	Lactic Acid.	10.40	0.00 - 16.40	ug/mgCR
8.	b-OH-Butyric Acid	4.72	0.00 - 9.90	ug/mgCR
8.	b-OH-Butyric Acid	4.72	0.00 - 9.90	ug/mgCR
9.	Glucose (OA)	0.4	0.1 - 1.1	mmol/L
9.	Glucose (OA)	0.4	0.1 - 1.1	mmol/L

CITRIC ACID CYCLE Metabolites.
(B Comp., CoQ10, Amino Acids, Mg)

CITRIC ACID CYCLE Metabolites.
(B Comp., CoQ10, Amino Acids, Mg)

10.	Citric Acid.	222.3	56.0 - 987.0	ug/mgCR
10.	Citric Acid.	222.3	56.0 - 987.0	ug/mgCR
11.	cis-Aconitic Acid.	20.5	18.0 - 78.0	ug/mgCR
11.	cis-Aconitic Acid.	20.5	18.0 - 78.0	ug/mgCR
12.	Isocitric Acid.	60.8	35.0 - 143.0	ug/mgCR
12.	Isocitric Acid.	60.8	35.0 - 143.0	ug/mgCR
13.	a-Ketoglutaric Acid.	9.80	0.00 - 35.00	ug/mgCR
13.	a-Ketoglutaric Acid.	9.80	0.00 - 35.00	ug/mgCR
14.	Succinic Acid	2.72	1.10 - 20.90	ug/mgCR
14.	Succinic Acid	2.72	1.10 - 20.90	ug/mgCR
15.	Fumaric Acid.	1.17	1.10 - 1.35	ug/mgCR
15.	Fumaric Acid.	1.17	1.10 - 1.35	ug/mgCR
16.	Malic Acid.	0.62	0.00 - 3.10	ug/mgCR
16.	Malic Acid.	0.62	0.00 - 3.10	ug/mgCR
17.	b-OH-b-Methylglutaric Acid	2.64	0.00 - 5.10	ug/mgCR
17.	b-OH-b-Methylglutaric Acid	2.64	0.00 - 5.10	ug/mgCR

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B-Complex Vitamins & Amino Acid Markers
(B1, B2, B3, B5, B6, Biotin)

B-Complex Vitamins & Amino Acid Markers
(B1, B2, B3, B5, B6, Biotin)

18.	a-Ketoisovaleric Acid	<dl		0.00 - 0.49	ug/mgCR
18.	a-Ketoisovaleric Acid	<dl		0.00 - 0.49	ug/mgCR
19.	a-Ketoisocaproic Acid	<dl		0.00 - 0.52	ug/mgCR
19.	a-Ketoisocaproic Acid	<dl		0.00 - 0.52	ug/mgCR
20.	a-Keto-b-Methylvaleric Acid	<dl		0.00 - 1.10	ug/mgCR
20.	a-Keto-b-Methylvaleric Acid	<dl		0.00 - 1.10	ug/mgCR
21.	Xanthurenic Acid	*0.58*	*H	0.0 - 0.5	ug/mgCR
21.	Xanthurenic Acid	*0.58*	*H	0.0 - 0.5	ug/mgCR
22.	beta-Hydroxyisovaleric Acid	7.95		0.00 - 11.50	ug/mgCR
22.	beta-Hydroxyisovaleric Acid	7.95		0.00 - 11.50	ug/mgCR

METHYLATION COFACTORS
(B12, Folate)

METHYLATION COFACTORS
(B12, Folate)

23.	Methylmalonic Acid.	1.77		0.00 - 2.30	ug/mgCR
23.	Methylmalonic Acid.	1.77		0.00 - 2.30	ug/mgCR
24.	Formiminoglutamic Acid **	1.4		0.0 - 2.2	ug/mgCR
24.	Formiminoglutamic Acid **	1.4		0.0 - 2.2	ug/mgCR

Cell Regulation Markers

NEUROTRANSMITTER METABOLISM
(Tyrosine, Tryptophan, B6, Antioxidants)

Cell Regulation Markers

NEUROTRANSMITTER METABOLISM
(Tyrosine, Tryptophan, B6, Antioxidants)

25.	Homovanillic Acid (HVA)	3.66		1.40 - 7.60	ug/mgCR
25.	Homovanillic Acid (HVA)	3.66		1.40 - 7.60	ug/mgCR
26.	Vanillylmandelic Acid (VMA)	2.53		1.20 - 5.30	ug/mgCR
26.	Vanillylmandelic Acid (VMA)	2.53		1.20 - 5.30	ug/mgCR
27.	5-Hydroxyindoleacetic Acid (5HIAA)	5.82		1.60 - 9.80	ug/mgCR
27.	5-Hydroxyindoleacetic Acid (5HIAA)	5.82		1.60 - 9.80	ug/mgCR
28.	Kynurenic Acid.	*1.65*	*H	0.0 - 1.5	ug/mgCR
28.	Kynurenic Acid.	*1.65*	*H	0.0 - 1.5	ug/mgCR
29.	Quinolinic Acid (OA)	5.11		0.00 - 5.80	ug/mgCR
29.	Quinolinic Acid (OA)	5.11		0.00 - 5.80	ug/mgCR
30.	Picolinic Acid	3.2		2.8 - 13.5	ug/mgCR
30.	Picolinic Acid	3.2		2.8 - 13.5	ug/mgCR
31.	Cortisol (OA)	434		166 - 507	nmol/L
31.	Cortisol (OA)	434		166 - 507	nmol/L

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Oxidative Damage/AntiOxidant Markers
(Vitamin C and Other Antioxidants)

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Oxidative Damage/AntiOxidant Markers
(Vitamin C and Other Antioxidants)

32.	ParaHydroxyphenyllactate	0.36	0.00 - 0.66	ug/mgCR
32.	ParaHydroxyphenyllactate	0.36	0.00 - 0.66	ug/mgCR
33.	8 OH-deoxyguanosine	3.4	0.0 - 7.6	ug/mgCR
33.	8 OH-deoxyguanosine	3.4	0.0 - 7.6	ug/mgCR

Toxicants and Detoxification

DETOXIFICATION INDICATORS
(Arg, NAC, Met, Mg, Antioxidants)

Toxicants and Detoxification

DETOXIFICATION INDICATORS
(Arg, NAC, Met, Mg, Antioxidants)

34.	2-Methylhippuric Acid	<dl	0.00 - 0.19	ug/mgCR
34.	2-Methylhippuric Acid	<dl	0.00 - 0.19	ug/mgCR
35.	Orotic Acid.	0.39	0.00 - 1.01	ug/mgCR
35.	Orotic Acid.	0.39	0.00 - 1.01	ug/mgCR
36.	Glucaric Acid.	*11.58* *H	0.00 - 10.70	ug/mgCR
36.	Glucaric Acid.	*11.58* *H	0.00 - 10.70	ug/mgCR
37.	a-OH-Butyric Acid	0.25	0.10 - 0.90	ug/mgCR
37.	a-OH-Butyric Acid	0.25	0.10 - 0.90	ug/mgCR
38.	Pyroglutamic Acid.	30.5	28.0 - 88.0	ug/mgCR
38.	Pyroglutamic Acid.	30.5	28.0 - 88.0	ug/mgCR

Compounds of Bacterial or Yeast/Fungal Origin

BACTERIAL DYSBIOSIS MARKERS.

Compounds of Bacterial or Yeast/Fungal Origin

BACTERIAL DYSBIOSIS MARKERS.

39.	Benzoate (OA)	1.84	0.00 - 9.30	ug/mgCR
39.	Benzoate (OA)	1.84	0.00 - 9.30	ug/mgCR
40.	Hippurate (OA)	*1157* *H	0.0 - 1070	ug/mgCR
40.	Hippurate (OA)	*1157* *H	0.0 - 1070	ug/mgCR
41.	Phenylacetate	<dl	0.0 - 0.2	ug/mgCR
41.	Phenylacetate	<dl	0.0 - 0.2	ug/mgCR
42.	Phenylpropionate	<dl	0.0 - 0.1	ug/mgCR
42.	Phenylpropionate	<dl	0.0 - 0.1	ug/mgCR
43.	ParaHydroxyBenzoate	0.8	0.0 - 1.8	ug/mgCR
43.	ParaHydroxyBenzoate	0.8	0.0 - 1.8	ug/mgCR
44.	p-HydroxyPhenylacetate	0.5	0.0 - 34.0	ug/mgCR
44.	p-HydroxyPhenylacetate	0.5	0.0 - 34.0	ug/mgCR
45.	Indoleacetic Acid	69.0	0.00 - 90.00	ug/mgCR
45.	Indoleacetic Acid	69.0	0.00 - 90.00	ug/mgCR
46.	Tricarballylate	0.89	0.00 - 1.41	ug/mgCR
46.	Tricarballylate	0.89	0.00 - 1.41	ug/mgCR

L. acidophilus/General Bacteria

L. acidophilus/General Bacteria

47.	D-Lactate	0.6	0.0 - 4.1	ug/mgCR
47.	D-Lactate	0.6	0.0 - 4.1	ug/mgCR

CLOSTRIDIAL SPECIES

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48.	Dihydroxyphenylpropionic Acid	<dl	0.00 - 0.05	ug/mgCR	
48.	Dihydroxyphenylpropionic Acid	<dl	0.00 - 0.05	ug/mgCR	
49.	4-Cresol	2.0	0.0 - 75.0	ug/mgCR	
49.	4-Cresol	2.0	0.0 - 75.0	ug/mgCR	
50.	3-OH-Proprionic Acid	70.7	0.0 - 208.0	ug/mgCR	
50.	3-OH-Proprionic Acid	70.7	0.0 - 208.0	ug/mgCR	

YEAST/FUNGAL DYSBIOSIS MARKERS.
YEAST/FUNGAL DYSBIOSIS MARKERS.

51.	Arabinitol	16.9	0.0 - 73.0	ug/mgCR	
51.	Arabinitol	16.9	0.0 - 73.0	ug/mgCR	
52.	Citramalic Acid	2.2	0.0 - 3.6	ug/mgCR	
52.	Citramalic Acid	2.2	0.0 - 3.6	ug/mgCR	
53.	Tartaric Acid.	4.1	0.0 - 7.0	ug/mgCR	
53.	Tartaric Acid.	4.1	0.0 - 7.0	ug/mgCR	

Oxalate Metabolites
Oxalate Metabolites

54.	Oxalic Acid	1.19	0.77 - 7.00	ug/mgCR	
54.	Oxalic Acid	1.19	0.77 - 7.00	ug/mgCR	
55.	Glyceric Acid	16.9	16.0 - 117.0	ug/mgCR	
55.	Glyceric Acid	16.9	16.0 - 117.0	ug/mgCR	
56.	Glycolic Acid	81.1	6.8 - 101.0	ug/mgCR	
56.	Glycolic Acid	81.1	6.8 - 101.0	ug/mgCR	

Nutritional Markers
Nutritional Markers

57.	Pyridoxic Acid (Vit B6)	*4.7* *L	5.0 - 34.0	ug/mgCR	
57.	Pyridoxic Acid (Vit B6)	*4.7* *L	5.0 - 34.0	ug/mgCR	
58.	Pantothenic Acid (Vit B5)	4.3	2.0 - 10.0	ug/mgCR	
58.	Pantothenic Acid (Vit B5)	4.3	2.0 - 10.0	ug/mgCR	
59.	Glutaric Acid (Vit B2) **	0.1	0.0 - 0.4	ug/mgCR	
59.	Glutaric Acid (Vit B2) **	0.1	0.0 - 0.4	ug/mgCR	
60.	Ascorbic Acid (Vit C)	21.1	10.0 - 200	ug/mgCR	
60.	Ascorbic Acid (Vit C)	21.1	10.0 - 200	ug/mgCR	
61.	CoEnzyme-Q10 (CoQ10) **	0.36	0.17 - 39.00	ug/mgCR	
61.	CoEnzyme-Q10 (CoQ10) **	0.36	0.17 - 39.00	ug/mgCR	
62.	N-Acetylcysteine (NAC)	0.19	0.10 - 0.28	ug/mgCR	
62.	N-Acetylcysteine (NAC)	0.19	0.10 - 0.28	ug/mgCR	
63.	Biotin (Vit H)	1.55	0.19 - 2.70	ug/mgCR	
63.	Biotin (Vit H)	1.55	0.19 - 2.70	ug/mgCR	
	Creatinine, Urine Spot.	17.6	8.0 - 19.0	mmol/L	
	Creatinine, Urine Spot.	17.6	8.0 - 19.0	mmol/L	

Results reported as <dl = Less than detectable limit

*** A high value for this marker may indicate a deficiency of this vitamin*

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Nutritional Guide

Nutrient	Adult Dose Range	Units	Clinician Notes
Vitamin-C	500.0	mg	
Vitamin-C	500.0	mg	
Vitamin-B1	15.0	mg	
Vitamin-B1	15.0	mg	
Vitamin-B2	70.0	mg	
Vitamin-B2	70.0	mg	
Vitamin-B3	275.0	mg	
Vitamin-B3	275.0	mg	
Vitamin-B5	55.0	mg	
Vitamin-B5	55.0	mg	
Vitamin-B6	150.0	mg	
Vitamin-B6	150.0	mg	
Vitamin-B12	450.0	ug	
Vitamin-B12	450.0	ug	
Chromium .	125.0	ug	
Chromium .	125.0	ug	
Magnesium .	600.0	mg	
Magnesium .	600.0	mg	
Coenzyme Q10.	300.0	mg	
Coenzyme Q10.	300.0	mg	
alpha Lipoic Acid.	125.0	mg	
alpha Lipoic Acid.	125.0	mg	
Calcium-D-glucurate.	1000.0	mg	
Calcium-D-glucurate.	1000.0	mg	
Acetyl-L-Carnitine.	600.0	mg	
Acetyl-L-Carnitine.	600.0	mg	
Probiotics (Multistain)	20.0	billion CFU	
Probiotics (Multistain)	20.0	billion CFU	
D-Lactate-free probiotics	5.0	billion CFU	
D-Lactate-free probiotics	5.0	billion CFU	

Disclaimer:

Supplement recommendations are based on the Organic Acid test results. The prescribing health practitioner must take into consideration the age, weight, sex, and pregnancy or lactation state. In addition, consider clinical state, medication regime, associated drug-nutrient depletion and allergies. The doses listed above are considered optimal, based on lab results and do not apply to specific disease conditions where doses may need to be altered. The vitamins, minerals or amino acids listed are elemental quantities. Use clinical discretion when choosing the right salt with the guidance of your compounding health professional. For example, Magnesium may be prescribed as a glycinate for its calming effect or threonate may be used for a Magnesium that crosses the blood-brain-barrier.

References:
Laboratory Evaluations for Integrative and Functional Medicine by Richard Lord.
J.Alexander Bralley; Textbook of Nutritional Medicine by Alan Gaby.

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Laboratory Comments

Ketone/FA Metabolites Comment

Organic acids provide functional markers for the metabolic effects of micronutrient adequacy, toxic exposure, neuroendocrine activity, intestinal bacterial and fungal overgrowth. Organic acid testing indicates the need for nutrients, diet modification, detoxification, antioxidant protection or further testing.

In a healthy state, organic acids are excreted in the urine at low concentrations. Low range results may be associated with hypometabolic compensatory states. Compensatory responses include hormonal secretions and cytokine responses that can slow or reverse deviations from median or normal physiologic states.

The Krebs cycle is a process of conversion of fats, carbohydrates and protein to mitochondrial energy, ATP.

Metabolic blocks in the Krebs cycle due to insufficient enzymes or cofactors will result in the elevation of organic acids that accumulate and spill into urine.

FATTY ACID METABOLISM:
Adipate, suberate, pimelate, Ethylmalonate and 2-methylsuccinate are organic compounds from fatty acid metabolism. Long chain fatty acids (LCFAs) undergo beta-oxidation in the mitochondria which is carnitine dependant. Dietary fat is broken down to produce free fatty acids, energy substrates using pathways that require carnitine and vitamin B2 (Riboflavin).
Low levels of Ethylmalonate with high adipate and suberate may be associated with carnitine deficient hypometabolic states where multiple amino acid catabolic pathways are restricted due to mitochondrial retraction.

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Carbohydrate Metabolism Comment

CARBOHYDRATE METABOLISM/GLYCOLYSIS:
Dietary carbohydrates are broken down into Glucose and other sugars where carbohydrate breakdown products, pyruvate and lactate are formed. Pyruvate enters the Krebs cycle via dehydrogenase enzymes which require vitamin B1 (thiamine), vitamin B2 (riboflavin), vitamin B3 (niacin), vitamin B5 (pantothenic acid), and lipoic Acid to function correctly. Review Vitamin B Levels in conjunction with Pyruvate and Lactate

levels.

In the absence of these nutrients, lactate builds up leading to lactic acidosis. Elevated pyruvate and lactate can indicate a need for lipoic acid.

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Cit Acid Cycle Metabs Comment

The Citric Acid Cycle is the pathway for energy released from food components and the source of anabolic molecules to support organ maintenance and neurological function. Therefore, the citric acid cycle serves both anabolic and catabolic functions representing the crossroads of food conversion and utilisation.

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B-Vitamins/Amino Acids Comment

B-COMPLEX VITAMIN MARKERS:
B-Complex Vitamin Markers are metabolic intermediates in the degradation of amino acids. When hepatic enzymes remove branched-chain amino acids, they form keto acids.
B-complex vitamins are essential for many in metabolic functions in the body used to extract energy from cellular health, remove toxins, and maintain the immune system.
B-Complex vitamin deficiencies produce symptoms associated with homocysteinemia effects or mitochondriopathy-associated symptoms which include periodic weakness, nausea, fatigue, attention deficit or Reye syndrome.

XANTHURENATE ELEVATED:
Xanthurenate is a by-product of tryptophan catabolism in the liver, arising from conversion of tryptophan to nicotinic acid which is Vitamin B6 dependent. Vitamin B6 deficiency leads to elevated excretion of the xanthurenate and kynurenate because of pyridoxal-5-phosphate-dependent step in this metabolic pathway called kynurenine pathway.

Elevations in Xanthurenate and Kynurenate are markers for Vitamin B6 (pyridoxal) deficiency. Assess blood Vitamin B6 levels.
Supplementation Recommendations:
Vitamin B6.

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B-Complex vitamin deficiencies produce symptoms associated with homocysteinemia effects or mitochondriopathy-associated symptoms which include periodic weakness, nausea, fatigue, attention deficit or Reye syndrome.

XANTHURENATE ELEVATED:
Xanthurenate is a by-product of tryptophan catabolism in the liver, arising from conversion of tryptophan to nicotinic acid which is Vitamin B6 dependent. Vitamin B6 deficiency leads to elevated excretion of the xanthurenate and kynurenate because of pyridoxal-5-phosphate-dependent step in this metabolic pathway called kynurenine pathway.

Elevations in Xanthurenate and Kynurenate are markers for Vitamin B6 (pyridoxal) deficiency. Assess blood Vitamin B6 levels.
Supplementation Recommendations:
Vitamin B6.

Neurotransmitter Metabolism Comment

5HIAA IS WITHIN RANGE:
5HIAA is the major metabolite of Serotonin.

KYNURENATE ELEVATED:
Elevations of this Tryptophan metabolite is consistent with vitamin B6 deficiency, possible inflammatory processes from chronic infections, interferon-gamma stimulated macrophages or excessive tryptophan supplementation (not 5HTP). Abnormally high levels can cause an increase in pain sensations and may, in multiple sclerosis (MS) patients, be a marker for an exacerbation period.
Kynurenate antagonises glutamate receptors and has a preferential affinity for NMDA receptor that governs synaptic plasticity learning and memory. Studies suggest that elevated Kynurenate represent conditions that are seen in schizophrenia, mediated by vitamin B6.
Kynurenate is a biochemical precursor to Quinolinate and found to be simultaneously elevated in inflammatory responses. Because Quinolinate is a powerful agonist of NMDA receptors and Kynurenate antagonises this effect, the relative amount of the two metabolites should relate to neuronal degradation.
Supplementation Recommendations:
Vitamin B6, Mg can help support metabolism of kynurenate.

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Detoxification/Toxicants Comment

OXIDATIVE DAMAGE AND ANTIOXIDANT MARKERS:
The assessment of protection from oxidant and ammonia challenge should be of priority when detoxification requirement is suspected. Oxidative stress has been associated with a variety of diseases like diabetes,

cancer, neurodegenerative disorders and aging.

DETOXIFICATION INDICATORS:
The organic acids of this group serve as biomarkers of detoxification status or biotransformation capacities, distinct parts of the detoxification system, providing insight about both exogenous toxin accumulation and endogenous detoxification responses.
Elevations in toxicant and detoxification markers reveal aspects of xenobiotic exposure, endogenous toxins and detoxification functions.

GLUCURATE ELEVATED:
Glucaric acid (glucarate) is a by-product of the predominant liver hepatic Phase I detoxification reactions involving cytochrome P450 oxidation of glucose to glucuronic acid, the substrate for phase II conjugation reactions. The best-known example is glucuronate conjugation of the haemoglobin degradation product bilirubin as a final preparation for urinary excretion.
Elevations in urinary glucarate specifically suggest exposure to pesticides, herbicides, fungicides, petrochemicals, alcohol, and drugs.
Consider a Liver Function Test to assess liver enzymes.

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Bacterial Dysbiosis Comment

HIPPURATE ELEVATED:
- Hepatic Phase II glycine conjugation of benzoate
- Commonly one of the most abundant compounds in normal urine

Causes:
When NO OTHER bacterial markers are high, only heavy dietary intake of benzoate is indicated (no clinical consequence).
- Benzoate production from intestinal bacterial overgrowth
- Confirm by examining other bacterial markers
- Excess intake of exogenous sodium benzoate (preserved, packaged food or cranberries)
- Malabsorption of phenylalanine due to low HCl in stomach
- Toluene exposure

Supplementation Recommendations:
- Glycine, 1000mg TID

Vitamin B5 (Pantothenic acid)500mg/day, Reduce toluene exposure.
Consider a Microbiome assessment to determine microbial overgrowth.

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BENJAMIN TRENGROVE
03-Oct-1989 Male

1 INTERNATIONAL PATIENT
MELBOURNE VIC 3000

-.I SCREEN
INTELLIGENT SCREENING
PO BOX 8441, ANGELO STREET
SOUTH PERTH WA 6151

LAB ID :	**3887019**
UR NO. :	**6190621**
Collection Date :	**23-Apr-2023**
Received Date:	**01-May-2023**

3887019

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Nutritional Markers Comment

PYRIDOXIC ACID (Vit B6) LOW:
Vitamin B6 and activated pyridoxyl 5-phosphate regulate inflammation, metabolism of PUFAs, haemoglobin synthesis and methylation. Deficiencies of Vit B6 is associated with seborrheic dermatitis, cheilosis, glossitis, microcytic anaemia and deficiency of neurotranmitters.
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